

Emergent Spacetime and Wave-Scale Readout from Anonymous Complex Zero Closure: Minimal Readout, an Infinite Constructive Solution, Logarithmic Spacetime Mapping, and Post-Mapping $\lambda\nu$ Invariance

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Abstract

This paper does not place space, time, wavelength, frequency, or velocity as foundational variables in advance. Starting only from an anonymous set of complex numbers identified under permutations and the complex square-zero closure

$$\sum_n z_n^2 = 0,$$

we investigate, in order of minimal cardinality, what can be read out from the inside.

First, a single complex number formally carries the quadratic form

$$z^2 = (a^2 - b^2) + 2iab,$$

but we distinguish this from the fact that, once the common complex scale is quotiented out as a gauge, no nontrivial relative quantity can be obtained from one element alone. For two nonzero complex numbers, imposing

$$z_1^2 + z_2^2 = 0$$

yields

$$z_2 = \pm iz_1,$$

so that the amplitude ratio

$$1$$

and the relative phase

$$\pm\pi/2$$

are fixed without any external measuring rod. For three elements, the squared quantities form a closed triangle in the complex plane, producing a cosine-rule-type constraint between relative amplitude ratios and relative phases. In general a continuous degree of freedom remains here, and we do not claim that a finite phase lattice arises automatically from square-zero closure alone.

Next, as the central construction of this paper, we give a countably infinite special solution built from an integer sequence

$$M_k$$

and phase cycles within each shell,

$$z_{k,n} = \frac{A}{M_k} \exp \left[i \left(\phi_k + \frac{\pi n}{M_k} \right) \right], \quad n = 0, \dots, M_k - 1.$$

Every shell closes exactly under the square-zero condition, and moreover, if

$$\sum_k \frac{1}{M_k} < \infty,$$

the infinite sum of squares converges absolutely. In particular

$$M_k = k(k+1)$$

is an explicit example that satisfies absolute convergence and

$$M_{k+1}/M_k \rightarrow 1$$

simultaneously.

In this special solution, while anonymity is preserved, a relative spatial scale

$$L_k \propto M_k$$

can be read from the shell amplitudes, and a relative period scale

$$P_k = M_k$$

can be read from the cycle order of the squared phase set. Therefore, before the mapping,

$$L_k/P_k$$

is constant, and setting

$$\nu_k^{(0)} := 1/P_k$$

gives

$$L_k \nu_k^{(0)} = \text{const}$$

.

Furthermore, since the continuous homomorphism that carries multiplicative scale ratios to additive relative coordinates is the logarithm up to a constant factor, we define, between two shells

$$j, k$$

,

$$\Delta X_{jk} = C_X \ln \frac{L_j}{L_k}, \quad \Delta T_{jk} = C_T \ln \frac{P_j}{P_k}.$$

After this spacetime readout, setting

$$\lambda_{jk} := |\Delta X_{jk}|, \quad \tau_{jk} := |\Delta T_{jk}|, \quad \nu_{jk} := \frac{1}{\tau_{jk}},$$

the special solution's

$$L_k \propto P_k$$

again yields

$$\lambda_{jk} \nu_{jk} = \frac{C_X}{C_T} = \text{const.}$$

Therefore, in this construction, the

$$\lambda\nu$$

-type reciprocal relation is preserved not only before the spacetime mapping but also after the actual mapping to additive spatial and temporal scales.

Finally, combining the two readouts, converted to the same units, into a single complex readout

$$W = X + iT,$$

the original squaring operation gives

$$dW^2 = dX^2 - dT^2 + 2i dX dT,$$

whose real part is a

$$1 + 1$$

-dimensional Lorentzian quadratic form. This paper does not claim that this construction is a unique derivation of real spacetime. What it shows is that the anonymous complex zero-closure system admits an explicit infinite special solution from which relative angles, relative scales, period scales, additive spacetime scales, a

$$\lambda\nu$$

-type invariant, and a Lorentzian signature can be read out in one consistent order.

Keywords: anonymous complex numbers, square-zero closure, readout, emergent spacetime, relative scale, logarithmic mapping, wavelength, frequency, Lorentzian metric, roots of unity

1. Problem Setting — Not Values, but “What Can Be Read”

In ordinary wave theory, one first places the spatial coordinate

$$x$$

, the time

$$t$$

, the wavelength

$$\lambda$$

, and the frequency

$$\nu$$

, and then describes a wave as

$$\exp\{i(kx - \omega t)\}.$$

This paper reverses the order.

What is given first is only a set of complex numbers and the closure of the sum of squares:

$$\mathcal{Z}_N = \{z_1, \dots, z_N\}/S_N,$$

$$Q(\mathcal{Z}_N) = \sum_{n=1}^N z_n^2 = 0. \tag{1}$$

The quotient by

$$S_N$$

is taken so that no complex number is given a physical name or order. Computational indices are mere memory locations, and readouts must not depend on permutations.

Equation (1) is also preserved under the common transformation by any

$$\gamma \in \mathbb{C}^\times$$

,

$$z_n \mapsto \gamma z_n,$$

under which

$$Q \mapsto \gamma^2 Q = 0.$$

Therefore equation (1) alone does not fix the common amplitude and the common phase. In this paper, quantities that depend on this common degree of freedom are not admitted as absolute readouts; differences, ratios, closure orders, and permutation-invariant symmetries are the candidates for internal readouts.

The question of this paper is simple.

Starting from an anonymous set of complex numbers with no background space and no background time, and importing no external measuring rod, at which stage can one construct relative quantities corresponding to angle, scale, period, space, time, wavelength, and frequency?

2. One Complex Number — Distinguishing the Formal Square Readout from the Gauge-Invariant Readout

Let a single complex number be

$$z = a + ib.$$

Its square is

$$z^2 = (a^2 - b^2) + 2iab. \tag{2}$$

Therefore, with respect to the fixed complex algebraic basis

$$(1, i)$$

,

- the difference quadratic form

$$a^2 - b^2$$

- the cross term

$$2ab$$

exist formally inside a single complex number.

However, if the common complex scale

$$z \mapsto \gamma z$$

, and in particular the common phase rotation, is quotiented out as a gauge, then the individual values of

$$a$$

and

$$b$$

, and the numerical value of

$$ab$$

, are in general not invariant. This paper therefore distinguishes

1. **that the square form exists algebraically**
2. **that it can be read as a relative quantity even after the gauge is quotiented out.**

With a single element there is nothing to compare with; under the full

$$\mathbb{C}^\times$$

gauge, all nonzero points belong to a single orbit. The strict sense of relative readout adopted in this paper therefore begins with two elements.

This distinction becomes important later. The Lorentzian quadratic form reappears at the final stage from the same square structure as equation (2), but at that point

$$X, T$$

are quantities already constructed as relative readouts.

3. Two Elements — The First Relative Angle without an External Angular Gauge

Suppose two nonzero elements

$$z_1, z_2$$

satisfy

$$z_1^2 + z_2^2 = 0. \tag{3}$$

Then

$$z_2^2 = -z_1^2,$$

and

$$\left(\frac{z_2}{z_1}\right)^2 = -1.$$

Therefore

$$\boxed{\frac{z_2}{z_1} = \pm i} \tag{4}$$

holds.

Equation (4) immediately gives

$$\boxed{\frac{|z_2|}{|z_1|} = 1} \tag{5}$$

and

$$\boxed{\arg z_2 - \arg z_1 \equiv \frac{\pi}{2} \pmod{\pi}} \tag{6}$$

Here the absolute phase is not determined. Rotating the two elements simultaneously preserves equation (3). However, **the right angle as a relative phase** does not disappear under common rotation.

Therefore the nontrivial zero closure of two elements is the minimal configuration that fixes the relative amplitude

$$1$$

and the relative angle

$$\pi/2$$

without first placing an external protractor.

Note that the

$$\pi/2$$

obtained here must not be confused with a claim that all phases of a general many-body system are quantized to four values. What equation (3) fixes is the relative angle intrinsic to the two-element closure.

4. Three Elements — Coupling of Relative Amplitudes and Relative Phases

Next consider three nonzero elements with

$$x^2 + y^2 + z^2 = 0. \tag{7}$$

Writing the squared quantities as

$$X = x^2, \quad Y = y^2, \quad Z = z^2,$$

we have

$$X + Y + Z = 0. \tag{8}$$

That is,

$$X, Y, Z$$

form a closed triangle in the complex plane. Since

$$Z = -(X + Y)$$

,

$$|Z|^2 = |X|^2 + |Y|^2 + 2|X||Y|\cos(\arg X - \arg Y).$$

Using

$$|X| = |x|^2$$

and

$$\arg X = 2\arg x$$

, we obtain

$$\boxed{\cos 2(\arg x - \arg y) = \frac{|z|^4 - |x|^4 - |y|^4}{2|x|^2|y|^2}} \tag{9}$$

The right-hand side of equation (9) consists only of relative amplitudes; the left-hand side consists only of relative phases. Thus, for the first time with three elements, the amplitude ratio, which collapsed to

$$1$$

for two elements, acquires a nontrivial shape degree of freedom, and this shape and the relative phase are bound by the same closure condition.

In this paper we call this the **conjugate-type amplitude-phase readout**. The word “conjugate-type” is used in the sense that fixing one constrains the other; it does not mean that canonically conjugate variables have been proved.

Also, equation (9) in general admits continuous shape parameters. Therefore, from

$$\sum_{j=1}^3 z_j^2 = 0$$

alone, one cannot say that relative phases are automatically restricted to a finite lattice such as

$$45^\circ$$

or

$$90^\circ$$

. Finite phase cycles are **explicitly realized as the structure of the solution** in the infinite special solution below.

5. A Constructive Special Solution with Countably Infinite Zero Closure

5.1 General shell construction

Take an integer sequence

$$M_k \in \mathbb{N}, \quad M_k \geq 2, \quad k = 1, 2, \dots$$

With

$$A \in \mathbb{C}^\times$$

a common scale and

$$\phi_k \in \mathbb{R}$$

a common phase for each shell, place

$$M_k$$

complex numbers for each

$$k$$

:

$$\boxed{z_{k,n} = \frac{A}{M_k} \exp \left[i \left(\phi_k + \frac{\pi n}{M_k} \right) \right]} \quad (n = 0, \dots, M_k - 1) \quad (10)$$

Squaring gives

$$z_{k,n}^2 = \frac{A^2 e^{2i\phi_k}}{M_k^2} \exp \left(\frac{2\pi i n}{M_k} \right). \quad (11)$$

Therefore, from the sum of

$$M_k$$

-th roots of unity,

$$\sum_{n=0}^{M_k-1} z_{k,n}^2 = \frac{A^2 e^{2i\phi_k}}{M_k^2} \sum_{n=0}^{M_k-1} \exp \left(\frac{2\pi i n}{M_k} \right) = 0. \quad (12)$$

Every shell closes exactly under the square-zero condition.

5.2 Absolute convergence of the infinite sum

The absolute value of each squared term is

$$|z_{k,n}^2| = \frac{|A|^2}{M_k^2},$$

so for the whole system

$$\sum_{k=1}^{\infty} \sum_{n=0}^{M_k-1} |z_{k,n}^2| = |A|^2 \sum_{k=1}^{\infty} \frac{1}{M_k}. \quad (13)$$

Hence, if

$$\sum_{k=1}^{\infty} \frac{1}{M_k} < \infty \quad (14)$$

the sum of squares converges absolutely. Therefore the sum does not change under any rearrangement of the terms as an anonymous set, and

$$\sum_{k,n} z_{k,n}^2 = 0 \quad (15)$$

holds exactly.

5.3 The minimal explicit example

In this paper we take

$$M_k = k(k+1) \quad (16)$$

as the representative example. Then

$$\sum_{k=1}^{\infty} \frac{1}{M_k} = \sum_{k=1}^{\infty} \frac{1}{k(k+1)} = 1. \quad (17)$$

Furthermore,

$$\frac{M_{k+1}}{M_k} = \frac{(k+1)(k+2)}{k(k+1)} = \frac{k+2}{k} \longrightarrow 1. \quad (18)$$

Therefore this sequence simultaneously satisfies

1. absolute convergence of the infinite sum of squares,
2. shell cycle order

$$M_k \rightarrow \infty$$

3. adjacent scale ratio

$$M_{k+1}/M_k \rightarrow 1$$

.

6. Re-identification of Shells from the Anonymous Set

In each shell of equation (10), the amplitudes of all elements coincide:

$$|z_{k,n}| = \frac{|A|}{M_k}. \quad (19)$$

If the

$$M_k$$

are chosen to be mutually distinct, the shells can be reconstructed from the anonymous set as input, through the **multiplicity of equal amplitudes**.

That is, letting

$$N(r_k)$$

be the number of elements with amplitude

$$r_k$$

, the special solution has

$$N(r_k) = M_k, \quad r_k = \frac{|A|}{M_k}. \quad (20)$$

Therefore

$$\boxed{r_k N(r_k) = |A|} \quad (21)$$

is common to all shells.

Even though

$$|A|$$

itself is a common scale and cannot be read, the ratio between shells

$$j, k$$

,

$$\frac{r_j}{r_k} = \frac{M_k}{M_j}, \quad (22)$$

can be read.

On the other hand, the squared phase set of each shell is

$$\left\{ \exp \left(2i\phi_k + \frac{2\pi i n}{M_k} \right) \right\}_{n=0}^{M_k-1},$$

which forms a regular

$$M_k$$

-gon. Its minimal cyclic generator is

$$U_k = \exp \left(\frac{2\pi i}{M_k} \right),$$

$$U_k^{M_k} = 1. \quad (23)$$

Therefore

$$M_k$$

is reconstructed identically both from the amplitude multiplicity and from the phase-cycle order. That “two independent readouts return the same integer

$$M_k$$

” is the central structure of this special solution.

7. Pre-Mapping Spatial and Period Scales and the $\lambda\nu$ Readout

7.1 Relative spatial scale

Absolute amplitudes are not read. However, the ratio of inverse amplitudes is

$$\frac{|z_k|^{-1}}{|z_j|^{-1}} = \frac{M_k}{M_j}.$$

We therefore read the space-type relative scale

$$L_k$$

obtained from amplitudes, up to a common constant, as

$$\boxed{L_k \propto M_k} \tag{24}$$

This is not a declaration of an absolute length such as “

$$M_k$$

meters”. What can be read is only the ratio

$$\boxed{\frac{L_j}{L_k} = \frac{M_j}{M_k}} \tag{25}$$

7.2 Relative period scale

The squared phases of the same shell return to the start after repeating the minimal rotation

$$2\pi/M_k$$

exactly

$$M_k$$

times. We therefore take the period-type readout to be

$$\boxed{P_k := M_k} \tag{26}$$

No second has been introduced here either.

$$P_k$$

is “the number of internal phase steps required for one full cycle”.

Its inverse is defined as the frequency-type readout,

$$\boxed{\nu_k^{(0)} := \frac{1}{P_k} = \frac{1}{M_k}} \tag{27}$$

7.3 The pre-mapping reciprocal relation

From equations (24) and (27), up to unit normalization,

$$\boxed{L_k \nu_k^{(0)} = \text{const}} \quad (28)$$

holds.

However, equation (28) alone is not sufficient.

$$L_k$$

and

$$P_k$$

are still multiplicative scale labels, and one must verify whether the same relation is preserved after mapping to ordinary additive spatial and temporal coordinates.

This is done in the next section.

8. Logarithmic Readout from Multiplicative Scales to Additive Spacetime

8.1 Why the logarithm

Relative scale ratios are multiplicative. For three shells

$$i, j, k$$

,

$$\frac{L_k}{L_i} = \frac{L_k}{L_j} \frac{L_j}{L_i}. \quad (29)$$

A continuous function

$$f$$

carrying this to additive relative coordinate differences

$$X_{ki} = X_{kj} + X_{ji}$$

must satisfy

$$f(ab) = f(a) + f(b).$$

Assuming continuity on the positive reals, its general form is

$$\boxed{f(r) = C \ln r} \quad (30)$$

Therefore the logarithm is not an arbitrarily chosen convenient transformation; it is **unique up to a constant factor as the continuous homomorphism that turns multiplicative scale ratios into additive relative distances.**

8.2 Spatial readout

Define the space-type relative interval between two shells

$$j, k$$

as

$$\boxed{\Delta X_{jk} := C_X \ln \frac{L_j}{L_k}} \quad (31)$$

In the special solution

$$L_k \propto M_k$$

, so

$$\Delta X_{jk} = C_X \ln \frac{M_j}{M_k}. \quad (32)$$

No absolute origin is needed, and

$$X \mapsto X + X_0$$

does not change the readout. In this sense

$$X$$

is a relative affine coordinate in one direction.

8.3 Time-scale readout

Ratios of periods are also multiplicative:

$$\frac{P_k}{P_i} = \frac{P_k}{P_j} \frac{P_j}{P_i}.$$

Therefore, by the same logic carrying period scales to additive time-scale differences,

$$\boxed{\Delta T_{jk} := C_T \ln \frac{P_j}{P_k}} \quad (33)$$

In the special solution

$$P_k = M_k$$

, so

$$\Delta T_{jk} = C_T \ln \frac{M_j}{M_k}. \quad (34)$$

A caution is needed here. What is directly obtained from the static anonymous set is the cyclic symmetry

$$C_{M_k}$$

of each shell and the period scale

$$P_k$$

. Equation (33) is a **relative time scale comparing the period scales of different clocks**; it is not a derivation of the arrow of time or of an infinite winding number. The phase states inside a shell give an

$$S^1$$

-type clock phase, but a complete time order

$$T \in \mathbb{R}$$

requires additional dynamics or an unwrap rule.

With this limitation made explicit, this paper examines the consistency of the relative metric between spatial and temporal scales.

9. $\lambda\nu = \text{const}$ **after the Spacetime Mapping**

This is the main verification of this paper.

Read the post-mapping spatial scale between two shells as

$$\boxed{\lambda_{jk} := |\Delta X_{jk}|} \quad (35)$$

Similarly, take the post-mapping period scale as

$$\boxed{\tau_{jk} := |\Delta T_{jk}|} \quad (36)$$

and define its inverse as

$$\boxed{\nu_{jk} := \frac{1}{\tau_{jk}}} \quad (37)$$

In the special solution

$$L_j/L_k = P_j/P_k = M_j/M_k$$

, so

$$\lambda_{jk} = C_X \left| \ln \frac{M_j}{M_k} \right|,$$

$$\tau_{jk} = C_T \left| \ln \frac{M_j}{M_k} \right|.$$

Hence

$$\boxed{\lambda_{jk}\nu_{jk} = \frac{C_X}{C_T} =: c_*} \quad (38)$$

Taking

$$C_X = C_T$$

in normalized units gives

$$\boxed{\lambda_{jk}\nu_{jk} = 1.} \quad (39)$$

When returning to dimensional units,

$$c_* = C_X/C_T$$

becomes the common constant converting spatial scales and temporal scales.

What matters is that equation (39) is not concluded merely from the pre-mapping identity

$$M_k \times \frac{1}{M_k} = 1.$$

Rather, **after** actually mapping

$$M \mapsto X = C_X \ln M,$$

$$M = P \mapsto T = C_T \ln P$$

to additive spacetime scales, we confirm that the

$$\lambda$$

and

$$\nu$$

redefined from intervals on that spacetime satisfy equation (38).

Therefore, in this special solution, the

$$\lambda\nu$$

-type relation is not destroyed by the spacetime readout; it commutes with the readout mapping.

10. The Lorentzian Quadratic Form

To convert the spatial and temporal scales to the same units, set

$$\tilde{T} := \frac{C_X}{C_T} T.$$

Below we again write

$$T$$

for simplicity of notation.

Combine the relative readouts into a single complex quantity

$$\boxed{W = X + iT} \tag{40}$$

Applying the same “complex squaring” as the starting point of this paper gives

$$\boxed{dW^2 = dX^2 - dT^2 + 2i dX dT} \tag{41}$$

Therefore the real part is

$$\boxed{\text{Re}(dW^2) = dX^2 - dT^2} \tag{42}$$

which carries the

$$1 + 1$$

-dimensional Lorentzian signature

$$(+, -)$$

.

The imaginary part is

$$\boxed{\text{Im}(dW^2) = 2 dX dT} \tag{43}$$

which is of the same form as the cross term appearing in the square of a single complex number,

$$(a + ib)^2 = a^2 - b^2 + 2iab.$$

Furthermore, in the special solution, for any two shells

$$j, k$$

the same scale ratio is read on the spatial side and on the temporal side, so

$$\frac{\Delta X_{jk}}{C_X} = \frac{\Delta T_{jk}}{C_T} = \ln \frac{M_j}{M_k}$$

holds. Here

$$C_X, C_T$$

are the constants defining the continuous homomorphisms from multiplicative scales to additive coordinates, and their signs correspond to the orientations of the respective readout axes. Since the underlying construction does not fix a positive direction of the axes, normalizing to the same absolute units

$$|C_X| = |C_T|$$

allows us to write

$$\Delta X_{jk} = \sigma \Delta T_{jk}, \quad \sigma = \pm 1.$$

Therefore, on equation (42), for every shell pair,

$$\boxed{ds_{jk}^2 = \Delta X_{jk}^2 - \Delta T_{jk}^2 = 0} \quad (44)$$

holds. That is, in this special solution it is not that some null-type branch exists: **every shell-pair interval is null**.

This is another expression of the same structure as

$$\lambda_{jk} \nu_{jk} = c_*$$

in Section 9. In this special solution,

$$L_k \propto P_k$$

simultaneously yields the post-mapping

$$\lambda \nu = \text{const}$$

and, in identical units,

$$ds^2 = 0.$$

Therefore the self-duality between “the spatial scale read from amplitudes” and “the period scale read from phase cycles”, the

$$\lambda \nu$$

-type invariant, and the null readout are three expressions of one identity structure in this special solution.

What is claimed here is not that all physics of real Lorentzian spacetime has been derived. Equation (42) is the algebraic result that a Lorentzian quadratic form naturally appears when the two relative scales constructed in this paper are fed back into the complex square readout. Also, the fact that all shell pairs become null is a property of this special solution; it does not mean that general square-zero-closure solutions are restricted to null readouts. Full Lorentz symmetry, causal order, the physical realization of boosts, and the extension to

$$3 + 1$$

dimensions remain separate problems.

11. The Asymptotic Continuum Limit

For the representative example

$$M_k = k(k+1)$$

, the adjacent differences of the spatial and temporal scales are

$$\Delta X_k = C_X \ln \frac{M_{k+1}}{M_k} = C_X \ln \frac{k+2}{k} \longrightarrow 0, \quad (45)$$

$$\Delta T_k = C_T \ln \frac{k+2}{k} \longrightarrow 0. \quad (46)$$

This refinement is not uniform; it becomes exponentially finer depending on the readout coordinate. Fixing a reference and taking

$$X_k = C_X \ln M_k = C_X \ln[k(k+1)],$$

we have for large

$$k$$

$$X_k \sim 2C_X \ln k, \quad k \sim \exp\left(\frac{X_k}{2C_X}\right).$$

On the other hand, equation (45) gives

$$\Delta X_k = C_X \ln\left(1 + \frac{2}{k}\right) \sim \frac{2C_X}{k},$$

so the local resolution at coordinate

$$X$$

is asymptotically

$$\boxed{\Delta X(X) \sim 2C_X \exp\left(-\frac{X}{2C_X}\right)} \quad (46a)$$

Therefore the continuum limit of this construction does not assume a uniform real lattice from the start; it is a **position-dependent asymptotic continuization** in which the discrete steps are exponentially refined as the logarithmic readout coordinate advances.

Also, the minimal step of the squared phases of each shell is

$$\Delta\theta_k = \frac{2\pi}{M_k} = \frac{2\pi}{k(k+1)} \longrightarrow 0. \quad (47)$$

Therefore, for large

$$k$$

,

1. the adjacent spacing of the spatial scale axis,
2. the adjacent spacing of the temporal scale axis,
3. the step of the clock phase

all become arbitrarily fine simultaneously.

In this paper we call this the **asymptotic local continuum limit**. The discrete set does not become the real line itself on a finite region from the start, but since the relative resolution goes to zero as

$$k \rightarrow \infty$$

, it has a local limit approximating a continuous metric.

In general, for integer sequences with the same properties in this construction it suffices to require at least

$$M_k \rightarrow \infty,$$

$$\sum_k \frac{1}{M_k} < \infty,$$

$$\frac{M_{k+1}}{M_k} \rightarrow 1. \quad (48)$$

For example, if

$$M_k$$

is an integerized

$$k^p$$

type with

$$p > 1$$

, one obtains a wide family of special solutions satisfying these three conditions.

12. Generalization of the Construction

The essence of equation (10) is not the particular

$$k(k+1)$$

. It is to place

$$M_k$$

phase-cycle points in shell

$$k$$

and to tie their common amplitude to

$$|z_k| \propto \frac{1}{M_k}. \quad (49)$$

By this condition, the same integer

$$M_k$$

appears both in

- the inverse scale obtained from amplitudes, and
- the cycle order obtained from the phase set.

The generalized sufficient condition is therefore

$$\boxed{\text{inverse-amplitude scale} \propto \text{phase-cycle order}} \quad (50)$$

If equation (50) holds, the multiplicative relative spatial scale and the relative period scale have the same ratio, so that even after the logarithmic mapping

$$\lambda\nu = c_*$$

is preserved.

This is a sufficient condition for isomorphic constructions beyond the special solution of this paper. However, this paper does not prove that equation (50) is necessary, nor that equation (50) necessarily emerges from general solutions of anonymous zero closure.

13. Relation to Prior Work

Each mathematical component of this paper has known nearby research. However, the starting point differs from the order of readout adopted here.

Penrose's twistor theory is the representative research describing Minkowski spacetime and conformal geometry using complex projective geometry and null structures [1]. There, however, Minkowski spacetime, or its complexification and conformal structure, is already positioned as the theoretical background. This paper differs in its problem setting: it inputs no spacetime coordinates and constructs relative spacetime scales from the internal readout of anonymous complex zero closure.

Nottale's scale relativity introduces relativistic structure into scale transformations themselves and discusses Lorentz-type scale transformations using logarithmic scales and a generalization of the de Broglie relations [2]. The closeness to this paper lies in treating multiplicative scales in logarithmic coordinates. In contrast, this paper does not give physical lengths or spacetime resolutions from the start; it first reconstructs the comparative scales themselves from the amplitude multiplicities and phase-cycle orders of an anonymous complex set.

Englman and Yahalom derived, from analyticity, an exact reciprocal relation between the phase and the log modulus for time-dependent wave functions [3]. That work is an important precedent showing that the phase and the logarithmic amplitude of a complex quantity are not independent. However, in that work the time

$$t$$

is given from the start as the independent variable of the wave function

$$\psi(t)$$

. The aim of this paper is to read out period scales and spatial scales from an anonymous set without background time.

In the literature survey at the time of writing, no prior work was found that executes the chain

$$\text{anonymous complex set} \rightarrow \sum z_n^2 = 0 \rightarrow \text{minimal relative readout} \rightarrow \text{explicit infinite zero-closure solution} \rightarrow (L, P) \rightarrow (\lambda, \nu)$$

from the same starting point. However, this is not a complete proof of nonexistence in the literature; it carries the qualification **to the best of our knowledge**.

14. What Is Shown as Theorems and What Remains as Physical Interpretation

To avoid overstating the claims of this paper, we separate what is shown directly in mathematics from what is adopted as readout interpretation.

14.1 Shown directly

1. From the nontrivial two-element closure

$$z_1^2 + z_2^2 = 0$$

follows

$$z_2/z_1 = \pm i$$

2. From the three-element closure follows the amplitude–phase constraint of equation (9).
3. The shells of equation (10) close exactly under the square-zero condition for each

$$k$$

4. Under the condition

$$\sum 1/M_k < \infty$$

, the infinite sum of squares converges absolutely.

5. The same

$$M_k$$

can be read both from the multiplicity of shell amplitudes and from the cycle order of squared phases.

6. The continuous homomorphisms from the multiplicative group

$$\mathbb{R}_{>0}$$

to the additive group

$$\mathbb{R}$$

are constant multiples of

$$\ln$$

7. If

$$L_k \propto P_k$$

, then

$$\lambda\nu = \text{const}$$

of equation (38) holds also after the logarithmic spacetime mapping.

8. The real part of the square of

$$W = X + iT$$

is

$$dX^2 - dT^2$$

14.2 Adopted as readout interpretation

1. Calling the inverse-amplitude ratio a space-type scale

$$L$$

2. Calling the phase-cycle order a period-type scale

$$P$$

3. Interpreting

$$\ln L$$

and

$$\ln P$$

as relative spatial and relative temporal scales, respectively.

4. Reading the post-mapping spatial scale difference as a wavelength

$$\lambda$$

and the inverse of the temporal scale difference as a frequency

$$\nu$$

5. Reading

$$\text{Re}(dW^2)$$

as a candidate physical spacetime line element.

These are mutually consistent and close on the same special solution. However, their uniqueness, and their identity with real physics, are subjects of future verification.

15. Limitations and Next Verification Problems

The problems left unsolved by this paper are clear.

15.1 Not a general solution

Equation (10) is a constructive special solution with strong symmetry. It is not shown that general solutions of the anonymous complex zero closure

$$\sum_n z_n^2 = 0$$

necessarily have shell structure or equation (50).

15.2 Reducible shell by shell

Since each shell closes to zero on its own, the whole system is reducible as a direct sum of local closures. The stronger next problem is whether the same readout holds for irreducible infinite closures that cannot be decomposed into finite subsystems.

15.3 The arrow of time is not derived

What is obtained from the static set is phase cycles and period scales. The physical process unwrapping the clock phase from

$$S^1$$

to

$$\mathbb{R}$$

, the time order, and the causal direction are not derived in this paper.

15.4 The extension from 1 + 1 to 3 + 1 is unresolved

What is constructed in this paper is one spatial scale direction and one temporal scale direction. By which constraints three independent spatial directions are selected from an anonymous complex set is a separate problem.

15.5 No absolute values of physical constants are produced

The

$$c_*$$

of equation (38) is a conversion constant between two readout units. This paper does not identify its numerical value with the known speed of light

$$c$$

. Connecting to real wave propagation requires independent observation or dynamical verification.

These limitations do not damage the constructive existence proof of this paper. Rather, they sharpen the next falsifiable question:

In more general or irreducible anonymous zero closures as well, do the amplitude-derived relative scale and the phase-derived period scale factorize through the same internal parameter, and is the

$$\lambda\nu$$

invariance after the spacetime mapping still preserved?

16. Conclusion

This paper did not start from “what is spacetime”; it started from “what can be read without an external measuring rod”.

The two-element complex zero closure fixes, through

$$z_2 = \pm iz_1,$$

the first gauge-invariant relative amplitude and relative angle. In the three-element closure, the relative amplitudes and relative phases are bound within the same constraint through the closed triangle in the squared plane.

Furthermore, constructing the countably infinite special solution

$$z_{k,n} = \frac{A}{M_k} \exp \left[i \left(\phi_k + \frac{\pi n}{M_k} \right) \right],$$

exact zero closure holds in each shell, and under

$$\sum 1/M_k < \infty$$

the infinite sum also converges absolutely. In this set, while anonymity is preserved, the double readout

$$\text{inverse amplitude} \longleftrightarrow M_k \longleftrightarrow \text{phase-cycle order}$$

holds.

As a result, before the mapping, the space-type scale

$$L_k$$

and the period-type scale

$$P_k$$

satisfy

$$L_k \propto P_k,$$

and mapping both from multiplicative to additive scales by the logarithm gives

$$\Delta X = C_X \ln \frac{L_j}{L_k}, \quad \Delta T = C_T \ln \frac{P_j}{P_k}.$$

The most important result is that for the wavelength- and frequency-type quantities redefined **after this spacetime readout**,

$$\boxed{\lambda\nu = c_*}$$

is preserved.

Furthermore, re-applying the complex square readout to

$$W = X + iT$$

yields the Lorentzian quadratic form

$$\boxed{\text{Re}(dW^2) = dX^2 - dT^2}$$

Therefore this paper gives, for the anonymous complex zero-closure system, one closed construction path:

minimal relative readout $\rightarrow$ amplitude–phase coupling $\rightarrow$ infinite closure special solution $\rightarrow$ $\lambda\nu$ -type readout $\rightarrow (X, T)$ r

This is not a claim to have uniquely derived real spacetime. The central result of this paper is to have shown that within complex zero closure with no background spacetime, there exists at least one infinite special solution from which relations isomorphic to the ordinary wave and spacetime metric can be read out self-consistently to the end.

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Appendix A: Uniqueness of the Logarithmic Readout

Suppose a map

$$f$$

carrying positive scale ratios

$$r > 0$$

to additive coordinates satisfies

$$f(rs) = f(r) + f(s). \tag{A1}$$

Setting

$$r = e^u$$

and defining

$$g(u) := f(e^u),$$

we get

$$g(u + v) = g(u) + g(v). \tag{A2}$$

If continuity is required of

$$f$$

, hence of

$$g$$

, the only continuous solutions of Cauchy's additive equation are

$$g(u) = Cu.$$

Hence

$$\boxed{f(r) = C \ln r.} \tag{A3}$$

Therefore the logarithm used for

$$X, T$$

in this paper is unique up to a constant factor under the requirement of carrying the multiplication of scale ratios to the addition of coordinate differences.

Appendix B: The Three Conditions for the Representative Sequence $M_k = k(k + 1)$

For the representative sequence

$$M_k = k(k + 1),$$

B.1 Absolute convergence

$$\sum_{k=1}^{\infty} \frac{1}{M_k} = \sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+1} \right) = 1.$$

B.2 Unbounded resolution

$$M_k \rightarrow \infty.$$

Therefore the phase step within a shell

$$\frac{2\pi}{M_k} \rightarrow 0.$$

B.3 Local continuization of relative scales

$$\frac{M_{k+1}}{M_k} = 1 + \frac{2}{k} \rightarrow 1,$$

therefore

$$\ln \frac{M_{k+1}}{M_k} \rightarrow 0.$$

From the above,

$$M_k = k(k+1)$$

is one of the simplest explicit examples that simultaneously satisfies the absolute convergence of the infinite closure and the asymptotic refinement of the spatial, temporal, and phase scales.